Marius Pinard

Robotics Engineering Student

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Final-year engineering student in Interactive Robotic Systems with hands-on experience in autonomous mobile robotics, ROS 2, SLAM, 3D perception and Industry 4.0 integration. **Seeking a 6-month final-year internship from March to August 2027** in robotics, autonomous systems, computer vision or industrial robotics.

Experience

SmartFactory-KL e.V. **Apr. 2026 – Jul. 2026**
Robotics Engineering Intern – Autonomous Mobile Robots & Industry 4.0 *Kaiserslautern, Germany*

- Integrated a **ZED 2 stereo camera**, **RoboSense 3D LiDAR** and NVIDIA Jetson AGX Orin on a Neobotix ROX AMR using **ROS 2 Jazzy**; implemented lifelong 2D SLAM, evaluated 3D mapping and adapted Nav2 for person-aware obstacles.
- Integrated the AMR into the SmartFactory environment through **OPC UA**, exposing ROS 2 / Nav2 navigation functions as industrial skills and validating autonomous navigation on the physical robot.

TLSe Racing – Formula Student **2025 – Present**
Driverless Division – Path Planning *Toulouse, France*

- Develop the **path-planning** component of a vision-based autonomous navigation system within a multidisciplinary Formula Student team.

Projects

Division Drone – French Gendarmerie Drone Hackathon | GitHub **Oct. 2026**
1st Place – UPSSITECH Team

- Built a **drone fleet mission-planning prototype** during a 9-hour hackathon, combining geospatial mapping, priority zones, resource allocation and coverage optimization.
- Developed a mission simulation with **React, TypeScript and Leaflet**, supporting drone deployment, coverage analysis, mission constraints and dynamic replanning.

Collaborative Industry 4.0 Production Cell **Oct. 2026 – Feb. 2027**
UPSSITECH – Final-Year Large-Scale Engineering Project

- Developing a **reconfigurable Industry 4.0 production cell** integrating TIAgO, Yaskawa HC10 cobots, Montrac transport and ZED 2 perception through a ROS 2 Jazzy / Gazebo Harmonic digital twin.

Interactive Mobile Robot **2024 – 2025**
UPSSITECH – Year-Long Robotics Project

- Designed and programmed a mobile robotic platform with **manual and voice-based control** for navigation in an indoor environment with obstacles, from design through integration and testing.

Education

UPSSITECH – University of Toulouse **2024 – 2027**
Engineering Degree – Interactive Robotic Systems (SRI) *Toulouse, France*

CUPGE – UPSSITECH **2022 – 2024**
Preparatory Program for Engineering Studies *Toulouse, France*

Technical Skills

Programming & Software Engineering: Python, C++, C, Java, object-oriented programming, Git, Linux
Robotics & Control: ROS / ROS 2, kinematic and dynamic modelling, robot control, motion and trajectory planning, autonomous navigation
Perception, Localization & AI: image processing, 3D vision, probabilistic estimation, Kalman and particle filters, SLAM, Machine Learning, Deep Learning
Embedded & Real-Time Systems: concurrent and multithreaded programming, synchronization, real-time systems, sensors / actuators, system dependability

Languages

French: Native **English:** B2 – Upper-intermediate **Spanish:** Intermediate